

Exercise 1F: Multiple Choice Questions

Complete Solved Guide & Explanation

Notebook: Comprehensive Fundamentals of Integer Mathematics

Mathematical Core & Sign Rules Summary

This solutions guide is grounded in the algebraic laws of integers ($\mathbb{Z}$) and the standard order of operations. Recall the essential rules:

- **Absolute Value ($|x|$):** Denotes the numerical distance from zero; $|x| \geq 0$ for all $x \in \mathbb{Z}$.
- **Additive Inverse:** The number $-a$ is the additive inverse of a because $a + (-a) = 0$.
- **Successor and Predecessor:** For any integer a , its successor is $a + 1$ and its predecessor is $a - 1$.
- **Signed Multiplication and Division:** Like signs yield positive values; unlike signs yield negative values.
- **BODMAS / PEMDAS Order of Operations:** Brackets $\rightarrow$ Of $\rightarrow$ Division / Multiplication $\rightarrow$ Addition / Subtraction.

Part 1: Multiple Choice Questions (Page 21)

1. The absolute value of 5 is:

- (a) -5
- (b) $\frac{1}{5}$
- (c) 0
- (d) 5

Correct Option: (d)

Solution. The absolute value (or modulus) of a number represents its distance from 0 on the number line, which is always non-negative.

$$|5| = 5$$

Thus, the absolute value of 5 is 5. □

2. The additive inverse of -7 is:

- (a) 7
- (b) $\frac{1}{7}$

- (c) 1
- (d) 0

Correct Option: (a)

Solution. The additive inverse of an integer a is the value $-a$ which, when added to a , results in the additive identity (0).

$$(-7) + x = 0 \implies x = -(-7) = 7$$

Therefore, the additive inverse of -7 is 7. □

3. The sum of two integers is -8 . If one of them is 5, then the other is:

- (a) 3
- (b) 13
- (c) -3
- (d) -13

Correct Option: (d)

Solution. Let the unknown integer be x . We can set up the linear equation:

$$x + 5 = -8$$

To isolate x , subtract 5 from both sides:

$$x = -8 - 5 = -13$$

Thus, the other integer is -13 . □

4. The successor of -41 is:

- (a) -42
- (b) 41
- (c) 40
- (d) -40

Correct Option: (d)

Solution. The successor of an integer a is defined as $a + 1$ (the number directly to its right on the number line).

$$\text{Successor} = -41 + 1 = -(41 - 1) = -40$$

Thus, the successor of -41 is -40 . □

5. The value of $(-7) \times 6 + (-7) \times 14$ is:

- (a) 42
- (b) -84
- (c) -140

(d) -196

Correct Option: (c)

Solution. By the **Distributive Property of Multiplication over Addition**, we have:

$$a \times b + a \times c = a \times (b + c)$$

Substitute $a = -7$, $b = 6$, and $c = 14$:

$$(-7) \times 6 + (-7) \times 14 = (-7) \times (6 + 14)$$

Simplify the expression inside the brackets:

$$6 + 14 = 20$$

Multiply:

$$(-7) \times 20 = -140$$

Thus, the expression evaluates to -140 . □

Part 2: Competency Focused Questions (Pages 22–23)

1. In a competition, 3 marks are given for every correct answer and (-2) marks are given for every incorrect answer and no marks for not attempting any question. Sachin scored 24 marks. If he got 14 correct answers, how many questions has he attempted incorrectly?

- (a) 10
- (b) 9
- (c) 8
- (d) 5

Correct Option: (b)

Solution. Let w represent the number of incorrect answers Sachin attempted.

- Marks scored from correct answers: $14 \times 3 = 42$ marks
- Marks lost from incorrect answers: $w \times (-2) = -2w$ marks
- Sachin's total score is 24.

Setting up the equation:

$$42 - 2w = 24$$

Rearranging to solve for $2w$:

$$2w = 42 - 24 = 18 \implies w = \frac{18}{2} = 9$$

Hence, Sachin attempted 9 questions incorrectly. □

2. A pair of integers whose product is -12 and there lies seven integers between them is:

- (a) $-2, 6$
- (b) $-12, 1$
- (c) $-3, 4$
- (d) $3, -4$

Correct Option: (a)

Solution. Let's analyze the properties of the options:

- i. For option (a), $-2 \times 6 = -12$ (Product is verified). The integers strictly between them on the number line are:

$$\{-1, 0, 1, 2, 3, 4, 5\} \quad (\text{Total of 7 integers})$$

This matches both conditions perfectly.

- ii. For option (b), $-12 \times 1 = -12$. The integers strictly between them are $0, -1, \dots, -11$ (Total of 11 integers).
- iii. For option (c), $-3 \times 4 = -12$. The integers strictly between them are $\{-2, -1, 0, 1, 2, 3\}$ (Total of 6 integers).

Thus, $(-2, 6)$ is the unique correct pair. □

3. Which of the following number sentences best describes the problem shown on the number line?

(Referencing the number line on Page 23, containing paths starting at 0)

- (a) $3 + (-2) + 8$
- (b) $3 - (-2) + 8$
- (c) $-3 + (-2) - 8$
- (d) $-3 + (-2) + 8$

Correct Option: (d)

Solution. Let us trace each movement shown by the directed arrows on the number line:

1. The first path starts at the origin 0 and goes 3 units to the left, terminating at -3 . This is represented by the integer -3 .
2. The second path starts at -3 and goes 2 more units to the left, terminating at -5 . This is represented by adding -2 .
3. The third path starts at -5 and moves 8 units to the right, terminating at $+3$. This is represented by adding $+8$.

The sum of these sequential translations is:

$$-3 + (-2) + 8 = 3$$

This translates exactly to option (d). □

4. Compare and fill in the box:

Successor of $((-8) \times ((-3) - (-5)))$ $\square$ predecessor of $((-12) \times ((36) + (-6)))$

- (a) $>$
- (b) $<$
- (c) $=$
- (d) Can't be determined

Correct Option: (a)

Solution. Let's evaluate the Left-Hand Side (L.H.S.) first:

$$\text{L.H.S. Value} = (-8) \times ((-3) - (-5)) = (-8) \times (-3 + 5) = (-8) \times 2 = -16$$

The successor of -16 is:

$$\text{L.H.S.} = -16 + 1 = -15$$

Now evaluate the Right-Hand Side (R.H.S.):

$$\text{R.H.S. Value} = (-12) \times ((36) + (-6)) = (-12) \times (36 - 6) = (-12) \times 30 = -360$$

The predecessor of -360 is:

$$\text{R.H.S.} = -360 - 1 = -361$$

Comparing the two results:

$$-15 > -361$$

Thus, the relational symbol in the box is $>$. $\square$

5. If we together add 199 negative integers, the resulting number will be:

- (a) Negative
- (b) Positive
- (c) Can't say
- (d) Data is insufficient

Correct Option: (a)

Solution. The sum of any set of negative integers is always negative. This is because:

$$a_i < 0 \implies \sum_{i=1}^n a_i < 0$$

Since each of the 199 integers is less than zero, their sum must also be strictly less than 0. Note that this differs from multiplication, where the sign depends on whether the count is even or odd. For addition, the sum remains negative regardless of whether the count (199) is odd or even. $\square$

6. Find x and y :

(i) The sum of two integers is 71. If one of them is -101 , then other integer is x .

(ii) The product of an integer and y is zero.

- (a) $x = 172, y = 0$
- (b) $x = 184, y = 1$
- (c) $x = 172, y = 1$
- (d) $x = 172, y = 2$

Correct Option: (a)

Solution. Let's solve each part:

i. The sum of -101 and x is 71 :

$$x + (-101) = 71 \implies x = 71 + 101 = 172$$

ii. The product of any integer a and y is 0 :

$$a \times y = 0$$

For this statement to hold generally for any integer a , y must equal 0 (by the Multiplicative Property of Zero).

Thus, $x = 172$ and $y = 0$, matching option (a). □

7. We have, $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times \frac{5}{9}$. If $^{\circ}\text{C} = -35$, then $^{\circ}\text{F}$ is equal to:

- (a) 31
- (b) -31
- (c) 95
- (d) -95

Correct Option: (b)

Solution. Substitute $^{\circ}\text{C} = -35$ into the given conversion formula:

$$-35 = (^{\circ}\text{F} - 32) \times \frac{5}{9}$$

Multiply both sides by the reciprocal fraction $\frac{9}{5}$ to isolate the term in parentheses:

$$-35 \times \frac{9}{5} = ^{\circ}\text{F} - 32$$

$$-7 \times 9 = ^{\circ}\text{F} - 32 \implies -63 = ^{\circ}\text{F} - 32$$

Add 32 to both sides:

$$^{\circ}\text{F} = -63 + 32 = -31$$

Thus, the temperature in Fahrenheit is -31°F . □

8. A shopkeeper earns a profit of Rs. 2 by selling a pen and incurs a loss of 50 paise per pencil and loss of 15 paise per eraser while selling pencils and erasers of old stock. On a particular day, he earns a profit of Rs. 10. If he sold 10 pens and the number of pencils and erasers he sold are in the ratio $7 : 10$, then the number of pencils and erasers he sold on the day is:

- (a) Number of pencils = 14, number of erasers = 20
- (b) Number of pencils = 12, number of erasers = 10
- (c) Number of pencils = 18, number of erasers = 10
- (d) Number of pencils = 12, number of erasers = 14

Correct Option: (a)

Solution. Let us work in paise to maintain integer calculations (1 Rupee = 100 paise):

- Profit per pen = Rs. 2 = 200 paise.
- Total profit from 10 pens = $10 \times 200 = 2000$ paise.
- Overall net profit achieved = Rs. 10 = 1000 paise.
- Total loss incurred on pencils and erasers = $2000 - 1000 = 1000$ paise.

The ratio of pencils to erasers sold is 7 : 10. Let the actual counts be:

$$\text{Pencils} = 7k, \quad \text{Erasers} = 10k \quad (\text{where } k \text{ is a positive integer})$$

Given losses: 50 paise per pencil, and 15 paise per eraser.

$$\text{Total Loss} = (7k \times 50) + (10k \times 15) = 1000 \text{ paise}$$

$$350k + 150k = 1000 \implies 500k = 1000 \implies k = 2$$

Using $k = 2$:

$$\text{Number of Pencils} = 7 \times 2 = 14$$

$$\text{Number of Erasers} = 10 \times 2 = 20$$

Thus, option (a) is correct. □

- 9. In a magic square, if each row, column and diagonal have the same sum, then the values of x and y respectively are:**

1	-10	0
x	-3	-2
-6	4	y

- (a) -7, -4
- (b) -4, -7
- (c) -4, -6
- (d) -6, -4

Correct Option: (b)

Solution. First, determine the required magic sum by calculating the sum of the completed Row 1:

$$\text{Magic Sum} = 1 + (-10) + 0 = -9$$

Since each row, column, and diagonal must sum to -9:

i. Sum of Column 1:

$$1 + x + (-6) = -9 \implies x - 5 = -9 \implies x = -4$$

ii. Sum of Column 3:

$$0 + (-2) + y = -9 \implies y - 2 = -9 \implies y = -7$$

Let us double check with other rows/diagonals to verify:

- Row 2: $-4 + (-3) + (-2) = -9$ (Verified!)
- Row 3: $-6 + 4 + (-7) = -9$ (Verified!)
- Diagonal (top-left to bottom-right): $1 + (-3) + y = 1 - 3 - 7 = -9$ (Verified!)
- Diagonal (top-right to bottom-left): $0 + (-3) + (-6) = -9$ (Verified!)

Therefore, $x = -4$ and $y = -7$. □

10. The additive inverse of $-28 + 12 + 42 - 53$ is:

- (a) 0
- (b) 27
- (c) $\frac{1}{27}$
- (d) $\frac{1}{-27}$

Correct Option: (b)

Solution. First, let us evaluate the expression:

$$-28 + 12 + 42 - 53 = (-28 - 53) + (12 + 42) = -81 + 54 = -27$$

The additive inverse of an integer A is $-A$.

$$\text{Additive Inverse of } -27 = -(-27) = 27$$

Thus, the additive inverse is 27. □

Comparison of Signs in Integer Arithmetic

Be careful not to confuse addition and multiplication sign rules:

- **Addition:** $(-5) + (-3) = -8$. (Sum of negatives is negative).
- **Multiplication:** $(-5) \times (-3) = 15$. (Product of negatives is positive).